

THERE’S MORE THAN ONE WAY: MODE COLLAPSE IN RLVR & MODEBENCH

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ABSTRACT

Reinforcement learning with verifiable rewards can improve success while concentrating the distribution of correct outputs. We introduce **MODEBENCH**, five executable domains whose validators identify which verified solution a model produces. These execution keys connect three forms of evidence: longitudinal training measurements, an idealized account of concentration, and controlled replay interventions. In *Graph* and *PantryPlan*, correct-key collision increases after *Dr.GRPO* and *GRPO* at all three tested model scales. Canonical replay reduces concentration relative to *Dr.GRPO* in those same domains, with explicit eligibility and sampling sensitivities. It also improves full-test success and sampled breadth, including alongside *MaxRL*. A key-weighting comparison tests why replay should balance distinct successes rather than their discovery frequency. Hosted evaluations show the relevance of the same measurement beyond these training runs, without identifying a training cause. Our theory proves concentration and banked-mode retention only under stated categorical mean-flow assumptions. Together, **MODEBENCH** and canonical replay provide a way to measure and revisit verified alternatives, with explicit limits on what the observations and theory establish.

1 INTRODUCTION

A model may learn to solve a problem more reliably while becoming less likely to propose a different correct solution. This matters whenever an agent samples alternatives, checks them, or revises a computation after feedback (Brown et al., 2024; Snell et al., 2024). Whether those alternatives help a later task is an empirical question. A prior measurement question is already necessary: which verified solutions does the model still produce?

Binary feedback in reinforcement learning with verifiable rewards (RLVR) (Guo et al., 2025) records success without distinguishing a familiar solution from another valid solution to the same prompt. Accuracy therefore leaves the allocation of probability among correct outputs unresolved. String diversity is insufficient as well: cosmetic variants can encode the same computation, and distinct computations can reach the same final answer. We need a reproducible identity for the successful output being counted.

MODEBENCH provides that identity through executable validators. Each accepted response receives a canonical key defining its task-specific mode. We study both the probability of finding a correct response and the distribution of keys conditional on correctness. This separates sampled breadth from conditional concentration, while retaining the dependence of finite-sample breadth on the number of successful draws.

The resulting evidence supports a concrete collapse story. In two domains with substantial initial breadth, *Graph* and *PantryPlan*, *Dr.GRPO* and *GRPO* increase correct-key concentration at three model scales. A conditional categorical analysis identifies an amplification mechanism for common correct modes. Canonical replay then uses the same execution identities as memory: one verified exemplar per retained key receives equal replay weight. *ReplayDr.GRPO* lowers measured concentration in the six corresponding model–domain comparisons. Controlled experiments with *Dr.GRPO* and *MaxRL* also test its effects on success and sampled breadth across all five domains.

These studies answer different questions about the same distribution. Checkpoint comparisons diagnose observed training changes; matched replay additions evaluate the implemented inter-

vention. Hosted deployments show that high correctness can coexist with concentrated verified outputs under their measured inference configurations. Their training histories are not controlled. We retain this distinction throughout: MODEBENCH supplies a common measurement, conditional theory explains a possible mechanism, and replay provides a tested intervention whose benefits and limits are visible together.

2 MODEBENCH: FROM CORRECTNESS TO VERIFIED ALTERNATIVES

For prompt x and specification s_x , the validator maps an output y to failure $\perp$ or a canonical key $V(y, s_x) \in \mathcal{C}_x$. Two accepted outputs count as the same mode exactly when their keys agree. Neither the full set $\mathcal{C}_x$ nor its size is supplied to training or the replay bank. The domain contracts determine the kind of alternative being measured:

Domain	Verified execution identity
Graph	A valid completed color vector.
Countdown	A normalized arithmetic computation tree.
Python	The executed return vector on the specified tests.
MathIR	A verified algebraic state trajectory.
PantryPlan	The ingredient support of a feasible plan.

These are task-defined output modes: configurations, computation routes, and tested program behaviors. They do not identify latent reasoning strategies or downstream usefulness. Cosmetic aliases merge under each validator’s rule; worked examples and complete prompts appear in Appendices D and E for all five task domains.

2.1 THREE COMPLEMENTARY MEASUREMENTS

Let $P = \Pr[V(Y, s_x) \neq \perp]$ and, when $P > 0$, let $q_c = \Pr[V(Y, s_x) = c \mid V(Y, s_x) \neq \perp]$. For K independent samples, D_K counts their distinct correct keys. The promptwise identities are

$$\underbrace{\Pr[D_K > 0]}_{\text{pass}@K} = 1 - (1 - P)^K, \quad \underbrace{\mathbb{E}[D_K]}_{\text{distinct}@K} = \sum_c [1 - (1 - Pq_c)^K]. \quad (1)$$

The first endpoint measures success probability; the second measures an expected number of verified keys. We use $K = 8$ for the training endpoints. Their difference, *extra modes*, counts verified keys beyond the first success. Both raw breadth and extra modes can increase because correctness increases; neither by itself establishes a more balanced q .

Conditional collision measures that allocation directly:

$$C(q) = \sum_c q_c^2, \quad B_2 = 2P - P^2C(q), \quad (2)$$

where $B_2 = \mathbb{E}[D_2]$. Thus C is the probability that two independent correct outputs share a key. Greater C means greater concentration. We call an observed increase in this conditional concentration *mode collapse*; finite samples do not establish literal extinction of unobserved modes. At fixed P , majorization toward a more concentrated q lowers expected breadth, but a change in C alone does not imply majorization or a change in total support (Appendix B).

3 CONCENTRATION DURING TRAINING AND IN HOSTED MODELS

3.1 LONGITUDINAL EVIDENCE FOR MODE COLLAPSE

We analyze saved initial and final outputs from eight-pass training runs on Qwen2.5-0.5B, Falcon3-1B, and Qwen2.5-3B. Graph and PantryPlan contain substantial initial sampled breadth; the other domains begin with at most .011 extra modes per prompt in the Dr.GRPO/GRPO precheck (Appendix H.2). Figure 1 therefore foregrounds these two informative domains while the appendix retains every domain, objective, and unavailable contrast.

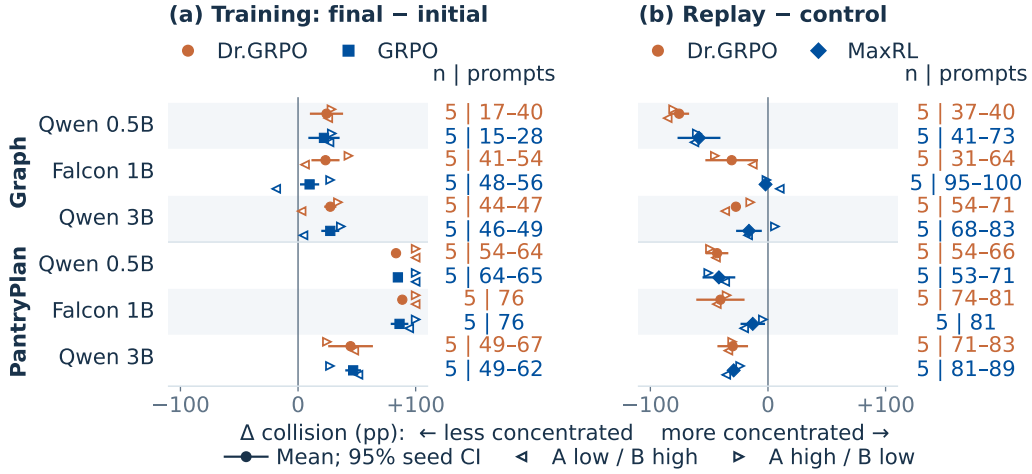


Figure 1: **Training concentrates verified outputs; canonical replay can mitigate it.** Left: final minus initial correct-key collision for Dr.GRPO and GRPO. Right: replay minus its matched terminal control, separately for Dr.GRPO and MaxRL. Graph and PantryPlan have substantial initial breadth. Dots and nominal 95% intervals average five seed contrasts; each contrast weights jointly eligible prompts equally. Smaller markers show the two disjoint-stream orientations; gutters give defined seed n and the eligible-prompt count range out of 128. The two panels have separate eligibility and are not one trajectory. Appendix H.1 reports all five domains, undefined cells, source assumptions, and correctness changes on the same eligible prompts.

All twelve Graph/Pantry Dr.GRPO/GRPO contrasts increase C , with primary nominal intervals above zero. Per-sample correctness increases on those same eligible prompts.

Both stream orientations agree in direction in eleven blocks; Falcon Graph with GRPO is the exception. Python initial outputs provide no jointly eligible Dr.GRPO/GRPO prompts, and several MathIR conditions are already observed at collision one. These are measured, heterogeneous training changes rather than a universal RLVR collapse law.

Eligibility and sampling. For $R \geq 2$ correct outputs with counts n_c , the estimator is $\hat{C} = \sum_c n_c(n_c - 1) / [R(R - 1)]$. Under fixed-policy iid sampling, $\mathbb{E}[\hat{C} | R] = C(q)$. We compare only prompts with a defined estimate in both conditions and report their coverage. Four saved eight-output groups reuse overlapping child seeds: under the audited runtime mapping they give eleven nominal streams. We select each stream’s earliest occurrence before inspecting its outcome and check separate five-versus-six-stream orientations. Shared randomness and selected eligibility make these empirical means descriptive; distinct seed IDs do not prove iid sampling. The original `pass@8` and `distinct@8` retain their intact groups. Appendix H.1 gives the full estimator proof, historical-runtime limits, and the explicitly timed retrospective cohort extension.

3.2 HOSTED OBSERVATIONS GIVE THE MEASUREMENT BROADER SCOPE

We evaluate seven hosted deployments on five domains and three levels, with 128 prompts and eight stateless draws per cell. Figure 2 shows per-response correctness and sampled breadth. Accurate Graph and MathIR outputs often revisit few keys, whereas PantryPlan exhibits broader outputs and substantial variation. These deployments establish the relevance of measuring verified alternatives; they do not identify the training process that produced their distributions.

The hosted collision summary pools correct pairs, weighting prompts by their observed correct-pair counts. Its population analogue weights by P^2 at a common draw budget, unlike the equal-prompt training contrast. The studies share the definition of C while retaining different aggregate estimands. A separate no-reasoning temperature sweep improves observed success and breadth through $T = 1.5$ on its measured profile; it shows that decoding can also change breadth, without identifying an optimal temperature in general (Appendix J.3).

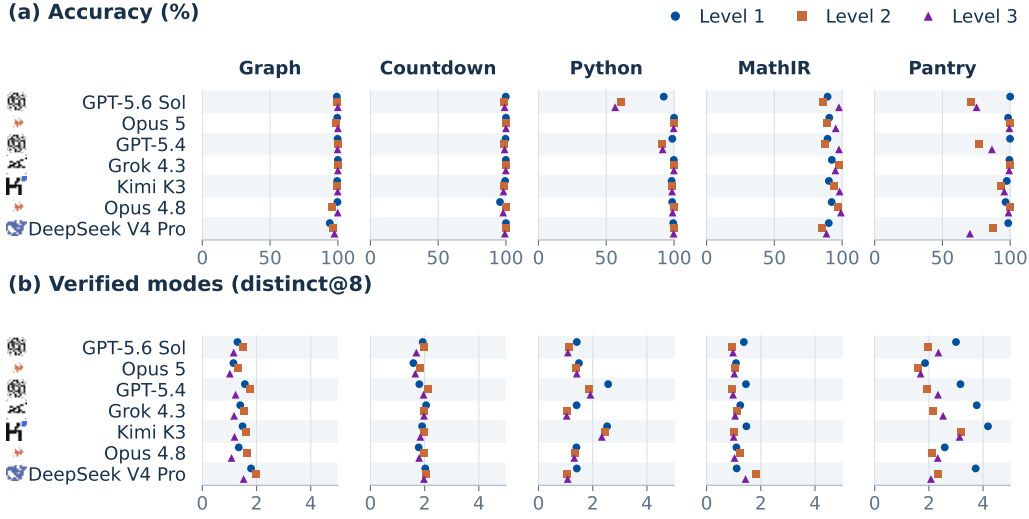


Figure 2: **Correctness leaves verified breadth unresolved in hosted deployments.** Columns show domains and markers distinguish Levels 1–3. Top: per-response accuracy; bottom: `distinct@8`, retaining all failed draws. A fixed formatting normalizer applies to every deployment. Opus 5 Python uses revised task wording without a system message; this display is not a common-prompt ranking. Appendix J reports original protocols, correct-pair collision, prompt comparisons, and uncertainty.

4 A CONDITIONAL MECHANISM AND CANONICAL REPLAY

Common correct modes appear in more on-policy updates than rare correct modes, despite identical binary rewards. To isolate this pressure, consider one prompt with at least two correct modes and one incorrect category, a finite category set, finite initial logits, and one independently optimized logit per canonical outcome. Assume common response-length normalization and Euclidean infinitesimal expected updates without a competing regularizer. For group size $G \geq 2$, at the on-policy point, the included GRPO/Dr.GRPO and binary MaxRL estimators induce a positive multiple of the correct-mass gradient. After a positive time change,

$$\frac{dq_c}{d\tau} = q_c(q_c - C), \quad \frac{dC}{d\tau} = 2 \text{Var}_{c \sim q}(q_c) \geq 0. \quad (3)$$

A unique initially largest correct mode attracts all conditional mass in this abstraction (Theorem C.3). The variance identity links the mechanism to the measured collision statistic. Independent canonical-outcome logits are an assumption: aggregation over response strings does not preserve neural gradient geometry, and the theorem does not certify AdamW/PPO or variable-length GRPO trajectories.

4.1 USE VERIFIED IDENTITY AS MEMORY

ReplayMaxRL combines MaxRL’s sampling-aware fresh objective (Tajwar et al., 2026) with an initially empty bank $\mathcal{B}_x$ per training prompt. Each admitted canonical key retains one policy-generated verified exemplar e_c ; repeated discoveries do not add entries. A deterministic schedule revisits nonempty banks, assigning equal weight to retained keys:

$$L_{\text{replay}}(x) = -\frac{1}{|\mathcal{B}_x|} \sum_{c \in \mathcal{B}_x} \frac{\log \pi_\theta(e_c | x)}{|e_c|}. \quad (4)$$

No gold solutions or catalogue of all valid modes is required. The bank contains training-prompt exemplars only; held-out evaluation measures a policy change rather than retrieval. Replacing MaxRL with Dr.GRPO yields ReplayDr.GRPO. Replay remains outside the fresh reward group. All-equal fresh rewards have zero centered task advantage, while stored targets can still supply a signal (Appendix F).

For the equal-weight categorical surrogate, bank mass $P_{\mathcal{B}}$ and its conditional allocation $q_{\mathcal{B}}$ satisfy

$$-\frac{1}{|\mathcal{B}|} \sum_{c \in \mathcal{B}} \log p_c = -\log P_{\mathcal{B}} + \log |\mathcal{B}| + \text{KL}(U_{\mathcal{B}} \| q_{\mathcal{B}}). \quad (5)$$

This identity explains the design: increase banked mass and oppose imbalance within the bank. The actual token-normalized exemplar loss is not an exact key-probability objective. Under the stated categorical flow, a fixed bank and constant positive replay dose yield a probability floor for banked modes; unbanked modes may vanish (Theorem C.6). Neither that floor nor a bounded exemplar surrogate establishes held-out transfer or visibility in eight draws. The controlled experiments test those observable outcomes separately from the conditional mathematical guarantees.

5 CONTROLLED REPLAY COMPARISONS

The central factorial crosses fresh objective (Dr.GRPO or MaxRL) with canonical replay off or on, at all three scales and five Level-1 domains. Each run trains for eight passes with five registered seeds. Controls execute the same bank and scoring code with zero replay gradient; realized banks and workloads can differ as policies diverge. Matched terminal comparisons retain 74 Dr.GRPO pairs and all 75 MaxRL pairs. Falcon Countdown has four admissible Dr.GRPO pairs; other primary blocks have five. Evaluation uses 128 held-out prompts and four fixed eight-output groups. Complete blocks receive nominal intervals; partial cohorts retain their exact counts without a five-seed interval. No missing or ambiguous endpoint is imputed (Appendices D.4 and L.1).

5.1 REPLAY IMPROVES SUCCESS AND SAMPLED BREADTH UNDER BOTH OBJECTIVES

Figure 3 presents the two matched replay additions on the same axes. ReplayDr.GRPO improves block-mean pass@8 and distinct@8 in every model-domain block; raw breadth increases in all 74 admitted seed pairs. ReplayMaxRL improves both endpoints in every 0.5B domain and their domain-averaged values at all scales. Falcon Graph’s raw-breadth effect is negative and inconclusive, making the domain variation visible alongside the overall benefit.

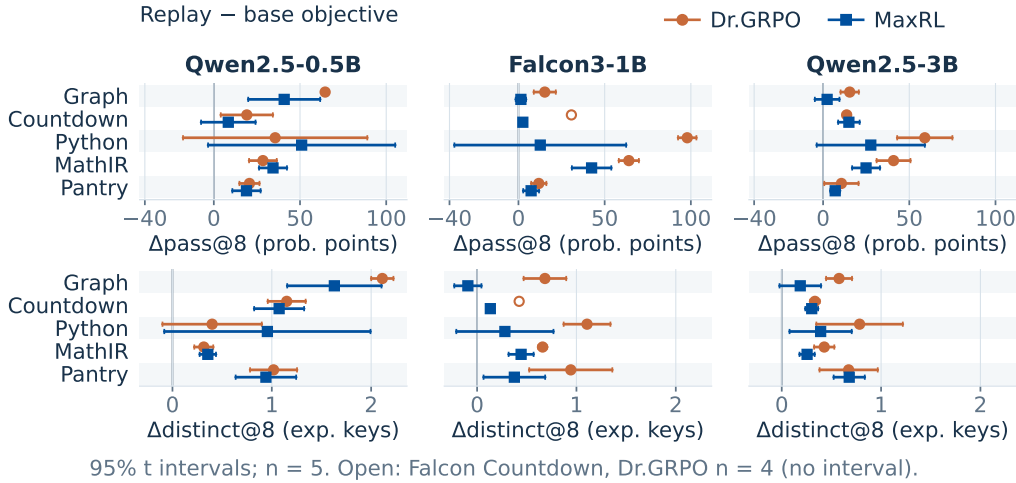


Figure 3: **Canonical replay complements both fresh objectives.** Each point compares replay with its matched no-replay control at pass eight, within domain and scale. The two endpoints have different units: success probability points and expected verified-key counts. All 75 MaxRL pairs are complete; Falcon Countdown uses four Dr.GRPO pairs and all other Dr.GRPO blocks use five. Nominal intervals accompany complete blocks only. Domain variation is retained; these comparisons identify the implemented replay addition rather than a universal benefit or factorial interaction.

At 3B, the completed MaxRL cohort has five seeds in every domain. Averaging domains equally within seed, replay raises pass@8 from .547 to .701 and distinct@8 from .744 to 1.107. All five domain means improve on both endpoints, while Graph and Python success intervals include zero. Full trajectories, absolute endpoints, and direct comparator results remain in Appendix G. UCPO and sparse RLEP-Dr are compared at 0.5B and 1B only; the fixed Semantic-MaxEnt comparator is defined in Appendix F.5.

5.2 CONDITIONAL CONCENTRATION IS A SEPARATE REPLAY OUTCOME

Figure 1’s right panel connects this intervention to the training diagnosis. ReplayDr.GRPO lowers C in all six Graph/Pantry comparisons, with negative primary intervals and agreement across both orientations. ReplayMaxRL has negative primary contrasts in those domains as well, but Falcon and 3B Graph change direction under one orientation. At 3B, MaxRL replay also lowers concentration in Countdown and Python; MathIR’s measured contrast is zero. These results support mitigation in demonstrated settings, not complete restoration of the initial distribution.

Correctness is not held fixed: the Graph/Pantry replay comparisons also have lower per-sample correctness on their jointly eligible prompts. This does not contradict higher full-test `pass@8`. They differ in population and in the event measured. Reporting conditional collision, full-test success, and sampled breadth separately avoids attributing a breadth gain solely to a more balanced q (Appendix H.1).

6 WHAT MATTERS, AND WHERE THE BENEFIT WEAKENS

6.1 BALANCING KEYS CONTRIBUTES BEYOND REUSING SUCCESSES

Uniform replay gives a rare retained key the same share of memory as a common one. A controlled weighting comparison replaces uniform key weights with weights proportional to fresh discovery frequency. Figure 4 shows the frozen Qwen2.5-0.5B comparison: uniform weighting adds .318 extra modes on average, with a nominal interval of $[-.272, .358]$, while the success contrast remains inconclusive. Graph, Countdown, and PantryPlan support the breadth contribution; Python and MathIR do not establish a uniform advantage. These results test the key-balancing choice, while full supporting scales and weighting details remain in Appendix G.6.

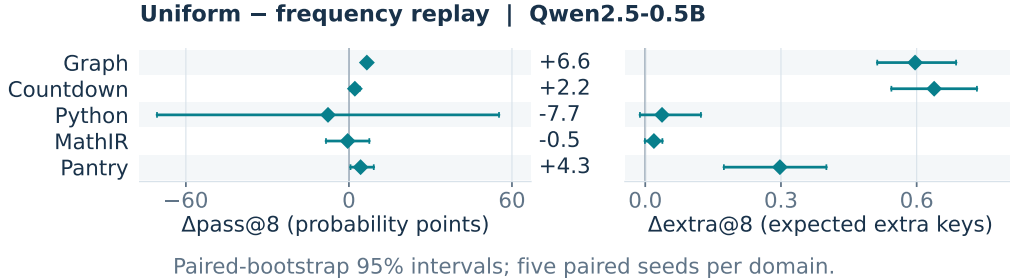


Figure 4: **Equal key weights can improve breadth beyond the first success.** Uniform minus fresh-frequency weighting of verified replay. Success and extra modes are separate endpoints; a positive extra-mode effect does not hold correctness fixed. The displayed model, domains, seeds, and uncertainty use the frozen weighting comparison. The domain variation and inconclusive success contrast limit the claim to the demonstrated breadth contribution.

6.2 REPLICATION WITHIN A SECOND TASK CONSTRUCTION

Level 2 retains validators and keys while changing problem structure. We train and compare all four arms again at Qwen2.5-0.5B. Graph, Countdown, Python, and MathIR have complete five-seed factorials; PantryPlan remains partial. Figure 5 shows *within-level* replay effects, holding the task population and prescribed protocol fixed for each comparison. Graph provides the clearest evidence. The four-domain aggregate improves in success and breadth under both fresh objectives, while the other complete domains’ individual intervals include zero.

Valid-mode-count matching applies to designated construction references, not identical native terminal tests. Cross-level changes also include parts of the prompt, syntax, and generation budget; the comparison is a replication under another construction rather than an isolated difficulty intervention (Appendix D.3). The conditional diagnostic adds a further boundary: sparse Level-2 Pantry comparisons show increased collision, from only one defined seed per objective. Neither those observations nor the positive Graph effects justify a universal concentration benefit.

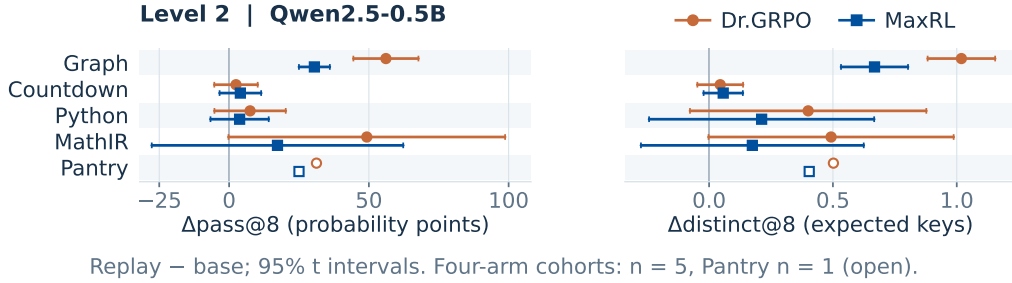


Figure 5: **Replay effects within Level 2.** Replay and control are compared on the same prompts and training seeds, with separate Dr.GRPO and MaxRL effects. Four domains have complete factorials; PantryPlan retains its partial counts and descriptive effects. Construction admission and cross-level absolute values remain in the appendix. Differences between levels combine task, population, and protocol changes, so they do not identify a causal effect of difficulty alone.

7 RELATED WORK AND DISCUSSION

Reasoning RL and RLHF can reduce output diversity, entropy, and high-budget coverage (Kirk et al., 2024; Dang et al., 2025; Cui et al., 2025; Yue et al., 2025). MODEBENCH measures executable output identities, whereas semantic clustering infers equivalence from language (Farquhar et al., 2024). Sampling-based selection motivates retaining proposal alternatives (Brown et al., 2024; Snell et al., 2024; Cobbe et al., 2021; Lightman et al., 2024); our experiments do not establish their downstream utility.

Outcome bonuses, set prediction, and distribution matching encourage broader support (Song et al., 2025; Chen et al., 2025; Li et al., 2025; 2026a;b; Lochab et al., 2026). MaxRL combines success objectives across sampling budgets (Tajwar et al., 2026); our factorial asks what memory adds. Success replay and experience replay address related reuse and forgetting questions (Zhang et al., 2025; Lin, 1992; Rolnick et al., 2019). Balancing execution keys connects the present memory design to behavioral archives in quality-diversity methods (Mouret & Clune, 2015; Ecoffet et al., 2021).

8 CONCLUSION

MODEBENCH reveals concentration that accuracy alone leaves unresolved. The controlled suite demonstrates training-induced concentration and replay mitigation in identifiable settings; hosted measurements establish broader relevance without identifying training causes. Conditional theory explains an amplification pressure and sufficient banked-mode protection, while the implemented replay experiment measures held-out outcomes.

A finite bank reaches only discovered and admitted modes. Fixed-exemplar scores improve centrally but retain a declining tail (Appendix H.4); exact key probabilities and survival of every stored mode remain unverified. The categorical guarantees do not certify neural optimization. Preserving verified alternatives is therefore a measured training objective here, with later search and adaptation benefits remaining questions for further experiments.